Akoizo Original Practice - Form 3 - Answer key

Division B | Roller Coaster | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) What is the minimum speed at the top of a loop with a 0.16 m path radius?

A. About 1.25 m/s

B. About 0.16 m/s

C. About 1.57 m/s

D. About 3.9 m/s

Correct: A

2. (2 points) What is the Size Score of a 60.0 x 30.0 x 80.0 cm device?

A. 0

B. 170

C. 90

D. -170

Correct: A

3. (2 points) Which change after impound is NOT allowed?

Unit: Size and construction.
Worked-case data: Option 1: 55 x 25 x 75 cm with a 22 cm loop. Option 2: 45 x 20 x 65 cm with a 15 cm loop. Compare Size + Loop points.
Separate investigation observations:
Home runs: Consistent 44.7 s.
After car ride: Runs 42.1 s; base slightly twisted.
After leveling: Runs 44.6 s.

A. Swapping the ball

B. Adjusting a slope

C. Moving the Finish Line

D. Leveling the device

Correct: C

4. (2 points) When does an elevator earn its 50 points?

A. When it is labeled

B. When it lifts the ball 5 cm

C. When the ball reaches the top track and continues on

D. When the counterweight moves

Correct: C

5. (4 points) Why can the ball never return to its starting height without an elevator?

answer: Friction and other losses remove energy, so the ball always has less energy than it started with. (4 points).

6. (4 points) Why record failure causes during testing?

Unit: Testing and competition day.
Worked-case data: Target 52 s. Practice run: 50.6 s. The calibration table says each screw turn adds 0.4 s. What should the team do?
Separate investigation observations:
Test week 1: Loop failed 3 of 10 runs.
Cause: Ball hit a raised joint just before the loop.
After sanding joint: Loop worked 10 of 10.

answer: They reveal the weakest part of the design so it can be fixed before competition. (4 points).

7. (4 points) Why calibrate each adjustment before competition?

Unit: Hitting the Target Time.
Worked-case data: Calibration: base path 34 s; each slope-screw turn adds 0.8 s; removable spiral adds 9 s. Plan for a Target Time of 46 s.
Separate investigation observations:
Runs at setting A: 41.8, 42.1, 41.9 s.
Runs at setting B: 44.6, 45.4, 44.8 s.
Target Time: 45 s.

answer: Knowing how much time each change adds lets the team hit a new Target Time quickly in 8 minutes. (4 points).

8. (4 points) Why are gradual loop entries and exits important?

Unit: Loops: physics and scoring.
Worked-case data: A loop has a 9.0 cm path radius. What is the ideal minimum release height above the loop bottom, and a practical target?
Separate investigation observations:
Release 24 cm: Ball fell from top of loop in 6 of 10 trials.
Release 28 cm: Ball completed loop 10 of 10.
Loop path radius: 9 cm.

answer: Sudden changes in curvature can make the ball bounce or jump the track. (4 points).

9. (4 points) Why are magnets and electronics excluded?

Unit: Rules and the four-part score.
Worked-case data: Target Time 45 s. A 52 x 26 x 72 cm device runs 44.6 s with both elevators and an 18.4 cm loop. Find the Run Score.
Separate investigation observations:
Design A: 15 cm loop, no elevators, size 30, time 215.
Design B: 20 cm loop, two elevators, size 15, time 210.
Target Time: 45 s for both.

answer: The rules prohibit magnets, electrical, and electronic devices anywhere in the roller coaster. (4 points).

10. (4 points) Why label every scoring feature?

Unit: Size and construction.
Worked-case data: Option 1: 55 x 25 x 75 cm with a 22 cm loop. Option 2: 45 x 20 x 65 cm with a 15 cm loop. Compare Size + Loop points.
Separate investigation observations:
Home runs: Consistent 44.7 s.
After car ride: Runs 42.1 s; base slightly twisted.
After leveling: Runs 44.6 s.

answer: Unlabeled elevators and loops may not be scored; labels make measuring and scoring clear. (4 points).

11. (6 points) Solve this case and explain your reasoning: A second-class lever has IMA 1.6. It drives a movable pulley inverted to give IMA 0.5. Is the combination legal, and how far does the ball rise if the effort moves 8 cm?

reasoning: Combined IMA = 1.6 x 0.5 = 0.8, which is less than 1.
Load distance = effort distance ÷ IMA = 8 ÷ 0.8 = 10 cm.
The ball rises exactly 10 cm, the minimum; add a small margin. (3 points).

conclusion: Legal with IMA 0.8; raise the effort travel slightly so the ball rises more than 10 cm. (3 points).

12. (2 points) What best explains the difference?

Supplied synthetic observations:
Predicted speed: 2.4 m/s at the bottom of a 0.40 m drop.
Measured speed: 2.0 m/s.
Track: Three joints along the drop.

A. Gravity was weaker

B. The ball gained mass

C. Energy losses from friction and the joints

Correct: C

13. (2 points) Define “Scorable run” in the context of Roller Coaster.

answer: Declared run that may count for scoring. (2 points).

14. (2 points) Define “Target Time” in the context of Roller Coaster.

answer: 30-60 s, revealed just before setup. (2 points).

15. (2 points) Define “Minimum top speed” in the context of Roller Coaster.

answer: v = sqrt (g r) at the top of a loop. (2 points).

16. (2 points) Define “Time Score” in the context of Roller Coaster.

answer: 5 points per full second up to the Target Time, minus 5 per second past it. (2 points).

17. (2 points) Define “Start Line” in the context of Roller Coaster.

answer: Labeled line the ball starts behind. (2 points).

18. (2 points) Define “Type 2 elevator” in the context of Roller Coaster.

answer: Second-class lever + pulley, IMA < 1. (2 points).